higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

PROFESSIONAL ENGINEERING PRACTICE
NQF LEVEL 4

13 NOVEMBER 2019

This marking guideline consists of 5 pages.

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QUESTION 1

1.1
1.1.1 D
1.1.2 D
1.1.3 A
1.1.4 B
1.1.5 C

(5 × 1) (5)

1.2
- Flammability
- Oxidising abilities
- Corrosiveness
- Ability to explode (4)

1.3 As manager, I will ensure that…
- workers receive all the benefits that the company offers.
- promotions in the workshop are fair.
- employees are informed about information and decisions taken by management
- employees have access to all policies and procedures.
- when employees do well, to acknowledge their good efforts.
- all employees are paid fairly for the work that they do.
- take enough breaks such as lunch and tea times.
- workplace is comfortable.
- the employees have the correct tools for their jobs.
- employees will state their training needs, and arrange training for them.

(Any 7 × 1) (7)

1.4
- Regular traffic jams on the roads, wasting time and people arriving late at their destinations
- Drivers become stressed, which may result in road rage
- Higher fuel consumption leads to more air pollution
- More wear and tear on vehicles
- Emergency vehicles may be prevented from reaching emergencies

(Any 4 × 1) (4)

1.5
- ECSA ensures that registered persons apply their knowledge to the advantage of humanity and the environment
- They ensure that registered persons do their work honestly according to set standards
- Ensuring that fellow persons respect each other
- Ensuring that registered persons honour the reputation of the profession
- Continuously improve the skills of registered persons
- Encourages excellence within the engineering profession (Any 5 × 1) (5)

[25]

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QUESTION 2

2.1 2.1.1 During this stage of the engineering design process, the project leader will create a team of experts and non-experts to generate as many concept solutions possible.✓ This is done by means of a brainstorming session where all concept solution ideas, even those that seem useless, are written on a whiteboard/flip-chart etc.✓ No criticism of any ideas is allowed.✓ Once the session is exhausted, the ideas will be ranked among those that are good, maybe's and not good.✓

2.1.2 When the final concept solution is chosen, the detail design of the concept, reflecting all specifications as determined by the customer and design team,✓ will be done. During this process, the team will do all calculations (strength, fatigue, stress concentrations, et cetera).✓ They will determine all materials required, as well as machining processes to be used.✓ Detail and assembly drawings indicating all dimensions and tolerances will be drawn.✓

2.1.3 When the detail design process is completed, a proper working prototype✓ of the design must be built according to the design details.✓ Once the prototype has been built to specification, a range of tests will be performed ✓ to determine whether the prototype adheres to all the specifications as determined by the team.✓

(3 × 4) (12)

2.2 2.2.1 9 (1)
2.2.2 7 (1)
2.2.3 56 (1)
2.2.4 118 (1)
2.2.5 Concept 1 (1)
2.2.6 Concept 1 received the highest rating factor of all three concepts
2.2.7 for reliability (1)
The criterion total for safety for concept 2 is incorrect.✓ As the weighting factor for safety is 9 and the rating factor for safety for concept 2 is 5, the criterion total for safety for concept 2 should be 9 × 5 = 45✓ (2)

2.3 2.3.1 True
2.3.2 False
2.3.3 False
2.3.4 True
2.3.5 True

(5 × 1) (5)
[25]

QUESTION 3

3.1
3.1.1 Isometric
3.1.2 Thin
3.1.3 Freehand
3.1.4 Smaller
3.1.5 Tolerance

(5 × 1) (5)

3.2
3.2.1 The introduction tells the reader what the purpose of the report is✓ and what the reader can expect in the remainder of the report.✓

3.2.2 The conclusion describes the main findings✓ resulting from the investigation done.✓

3.2.3 The recommendations section describes the actions that the author of the report recommends for solving the problem✓ and it might also suggest changes and/or future work/maintenance to be done.✓

3.2.4 The appendices contain all documents that are too bulky and detailed to include in the main report,✓ such as graphs, calculations, tables, et cetera.✓

(4 × 2) (8)

3.3
- 'Turn work piece' is an action,✓ and should be placed in a rectangular box.✓
- 'Can the fault be rectified on the lathe?' is a question and is indicated in a rectangular box.✓ The rectangular box must be replaced with a diamond box.✓
- The line of action to 'scrap component' does not have an arrow.✓ An arrow should be added.✓
- 'Quality of packaging' is in a diamond box and should be in question form as it has 'Yes' and 'No' answers flowing from it.✓ Change to: 'Quality of packaging acceptable?'✓
- 'Redo packaging to required standard' does not have a line flowing from it to indicate the flow once the packaging has been rectified.✓ Include an arrow line to correct 'Quality of packaging' box.✓ (5 × 2) (10)

3.4
- Block diagrams
- Logical networks (2)

[25]

QUESTION 4

4.1	4.1.1	J		
	4.1.2	A		
	4.1.3	C		
	4.1.4	K		
	4.1.5	L		
	4.1.6	I		
	4.1.7	M		
	4.1.8	B		
	4.1.9	E		
	4.1.10	D		
			(10 × 1)	(10)

4.2 •Agile development
•Clean-room development (2)

4.3 •AND
•OR
•NOT (3)

4.4	4.4.1	Input device		
	4.4.2	Storage device		
	4.4.3	Storage device		
	4.4.4	Output device		
			(4 × 1)	(4)

4.5	4.5.1	•Bold •Italic	(2)
	4.5.2	Fitting and turning	(1)
	4.5.3	The enrolment numbers for Fitting and Turning declined from 80 to 65 to 44 over three years	(1)
	4.5.4	✓ ✓ ✓ ✓ = AVERAGE(F6:F9) or = average(F6:F9)	(2)
			[25]

TOTAL: 100

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